

Percussion Sequencer

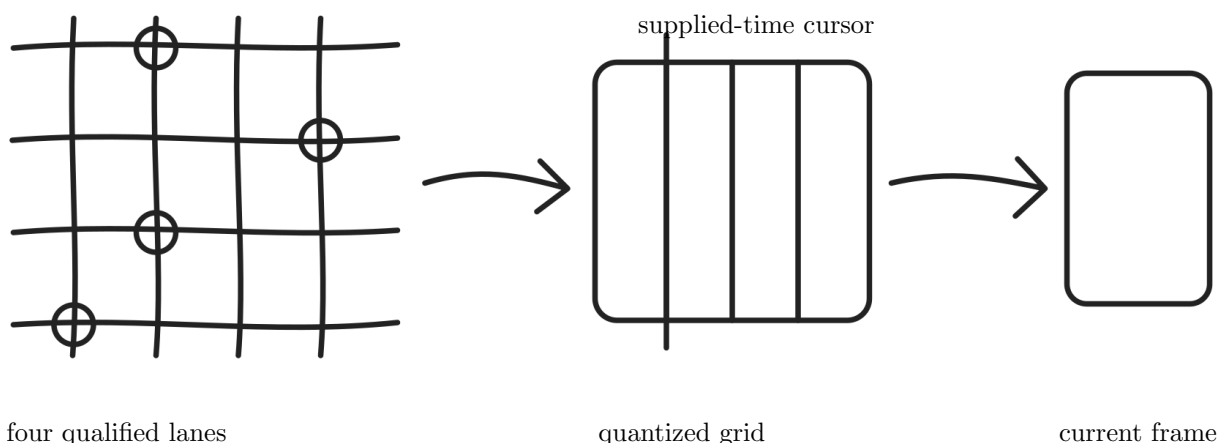
Group attacks, associate one envelope, replay bounded cues

Exact reference design; incoming conformance and E1 acceptance open

Lesson	039 — project	Time	180–240 minutes
Prerequisites	Lessons 037–038	Board	Mega 2560 reference fixture
Evidence	six LEDs, passive piezo, named test points	Status	integration verification active; bench open

Scope gate. Four C&K/Littelfuse RB-220-07A R contacts and one SparkFun SEN-12642 V10 define the reference circuit. A look-alike kit part is not interchangeable. Intensity is relative ADC evidence, not sound pressure, loudness, force, damage, speech, or a safety measurement.

1 From four lanes to one pattern



Alternative text: four hand-drawn lanes feed a bounded sixteenth-note grid. A supplied-time cursor selects only the current frame.

The first qualified attack freezes recording tempo and epoch. Up to one attack per lane enters an inclusive simultaneous window. One eligible completed acoustic interval supplies relative intensity to every group member; timeout supplies zero with explicit `AssociationTimeout` provenance. Only one closed group waits, so later attacks are visibly suppressed rather than secretly queued.

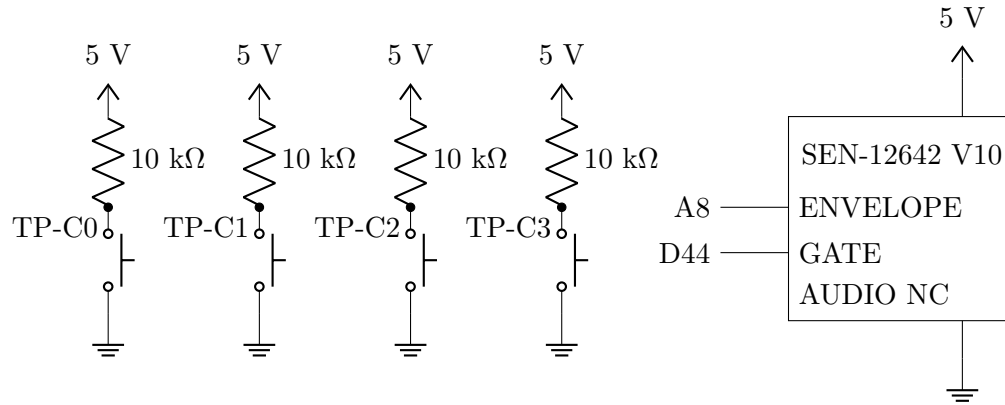
The pattern contains 4–16 sixteenth-note steps at 30–240 BPM:

$$\text{stepPeriodMs} = \left\lfloor \frac{60000}{4 \text{ tempoBpm}} \right\rfloor.$$

Nearest-step quantization rounds an exact half-step forward. Fixed storage holds 32 hits. A group that cannot fit is rejected atomically. `hit(index)` returns a copied hit with its association provenance.

2 Electrically authoritative input schematic

The input schematic below is authoritative for this named reference fixture only. Each contact has its own 10 k Ω external pull-up. The adapter therefore uses `Pull::None` and interprets closure as active low.



Four nonpolar rolling-ball contacts close TP-C0–TP-C3 to ground. The module and Mega share 5 V and ground; AUDIO is disconnected.

Path	Mega pin	Observation
TP-C0–TP-C3	D40–D43	high open; low when its reference contact closes
TP-E / TP-T	A8 / D44	envelope ADC and active-high gate
TP-BPM	A9	10 k Ω potentiometer wiper
surface evidence	D45–D48	accepted/current surface, each through 1 k Ω
step display	D49–D51	74HC595 data, clock, and latch
state/capture	D52/D53	ready/heartbeat and hardware-sample interval
cue	D6 / timer 2	identified passive piezo, bounded tone only
play/clear	D38/D39	buttons to ground using internal pull-ups

3 Observe, decide, actuate

The Mega adapter first updates four inputs and `ContactDynamics` policies, then the envelope and gate inputs and `AcousticEnvelope`. It copies only qualified attack bits, source statuses, and a completed acoustic record into `PercussionSequencerInput`. The engine owns no pin, timer, clock, endpoint, or dynamic storage.

Playback position derives from elapsed supplied time. Sparse calls skip to the frame current now and never replay crossed cues late. A one-digit display shows the active step. Accepted-hit LEDs latch for 180 ms and show current surfaces during playback. In reference-hardware mode only, D53 toggles on each coalesced 20 ms hardware sample; replay leaves it inactive. The strongest current hit selects 262, 330, 392, or 523 Hz; equal intensity selects the lowest lane. A frame without a hit has zero tone intent. That does not prove physical silence.

4 Predict, observe, interpret

1. Keep `ADK_LESSON039_USE_REFERENCE_HARDWARE=0`. Predict three replay hits followed by playback from step zero.
2. Observe D45–D48 for accepted/current surfaces. Jumping time must not produce a burst of missed LED or tone cues.

3. Observe D52 steady as ready outside playback and changing once per beat during playback, including empty pattern regions. D53 stays inactive in replay; on a qualified reference build it toggles on each coalesced 20 ms hardware sample.
4. Inject one failing source status in host replay. Predict retained host fault attribution while D52, D53, the display, surface LEDs, and piezo all become inactive.
5. At the simultaneous boundary predict inclusion; one tick later predict suppression while association waits.
6. Omit completion. Predict zero intensity with timeout provenance, not a claim that the room was quiet.

5 Qualified-build and shutdown checklist

Before selecting hardware mode, match all five incoming specimens to the named references, inspect unpowered wiring, and approve a bounded powered plan. Measure the actual 5 V rail and every pull-up; prove each worst-case closed current is no greater than the RB reference's 1 mA maximum. Measure ENVELOPE and GATE ranges against Mega limits. Complete per-pin, per-port, 74HC595-package, 5 V rail, and total-current arithmetic; the known direct LED and display bound is 70 mA, excluding module, pulls, passive piezo, and Mega baseline. Power the module and Mega together; never drive a module output into an unpowered Mega.

Resource acquisition and electrical safe state are separate evidence. Initialization status establishes claims. After shutdown, verify LEDs and piezo inactive and separately inspect TP-C0–TP-C3, TP-E, and TP-T. Remove USB for heat, odor, reset, unstable rail, or unexpected output. Rewire only while unpowered.

Physical acceptance remains open. Compilation, host replay, documentation, and an authoritative reference schematic do not prove the delivered specimens, currents, output ranges, closure orientation, sounder behavior, or safe shutdown on a bench. Record those observations only after the qualified fixture is measured.

6 Software evidence and sources

The project-engine tests exercise grouping and association boundaries, quantization, capacity, playback, source attribution, lifecycle, and the versioned fieldwise trace. Focused example tests exist for replay scheduling, frame-cue gating, and the hardware-free adapter path; their shared promotion gate remains integration work.

- C&K/Littelfuse RB-series formal datasheet, exact RB-220-07A R drawing and ratings.
- SparkFun SEN-12642 hardware V1.0 source at repository commit [a456d9e](#), product page, and hookup guide.
- Texas Instruments LMV324 primary datasheet.
- ADK sensor evidence report, testing contract, and safety model.

This untagged print edition has a searchable HTML alternative. It makes no PDF/UA or physical accessibility claim.